

Random Graph

Foundation of Data Science

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May 17th, 2019

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1. Nonuniform Models of Random Graphs
2. Growth Models
3. Small World Graphs

Nonuniform Models of Random Graphs

Nonuniform random graphs

- Until now, we studied the random graph $G(n, p)$ in which all vertices have the same distribution of expected degree, and the degree distribution is concentrated on its expected value.
- However, the graph in real world does not have this property and the degree distribution follows the **power law**, i.e., the number of vertices of degree d decays as $\frac{1}{d^\alpha}$ with some $\alpha > 0$.
- We now consider the fixed degree distribution $f(d)$ and consider the random graph G that is uniformly selected from the all set of graph with degree distribution $f(d)$.
- Since the edges are now not independent to each other, it is hard to analyze the random graph.
- However, some of the phase transition still can be studied in the nonuniform random graph.

Giant component in graphs with given degree distribution

- We study the phase transition of the giant component.
- The **Molloy-Reed** condition for having a giant component is

$$\sum_{i=0}^{\infty} i(i-2)\lambda_i > 0,$$

where λ_i is the fraction of vertices of degree i .

- The heuristic explanation of this condition is as follows: for each edge connection, the number of frontier increases by $i - 2$ when the vertex with degree i is connected. The term $i\lambda_i$ is proportional to the probability of reaching a degree i vertex.
- Thus, the above sum somehow implies the expected value of the increment of the number of frontier for each edge connection. If this is positive, then there will be a giant component.

Example: M-R condition on $G(n, p)$

- Recall that the degree distribution of $G(n, p)$ is given by

$$p_i = \binom{n}{i} p^i (1-p)^{n-i}.$$

- We will show that

$$\lim_{n \rightarrow \infty} \sum_{i=0}^n i(i-2)p_i = 0,$$

This implies there exists a phase transition at $p = \frac{1}{n}$, which is consistent to the result in the previous section.

$$\begin{aligned}
& \lim_{n \rightarrow \infty} \sum_{i=0}^n i(i-2) \binom{n}{i} \left(\frac{1}{n}\right)^i \left(\frac{n-1}{n}\right)^{n-i} \\
&= \lim_{n \rightarrow \infty} \left(\frac{n-1}{n}\right)^n \sum_{i=0}^n i(i-2) \frac{n(n-1) \cdots (n-i+1)}{i! n^i} \left(\frac{n-1}{n}\right)^{-i} \\
&= \frac{1}{e} \lim_{n \rightarrow \infty} \sum_{i=0}^n i(i-2) \frac{n(n-1) \cdots (n-i+1)}{i! n^i} \left(\frac{n}{n-1}\right)^i \\
&= \frac{1}{e} \lim_{n \rightarrow \infty} \sum_{i=0}^n \frac{i(i-2)}{i!} \frac{n(n-1) \cdots (n-i+1)}{(n-1)^i} \leq \sum_{i=0}^{\infty} \frac{i(i-2)}{i!}.
\end{aligned}$$

However, since

$$\sum_{i=0}^{\infty} \frac{i^2}{i!} = \sum_{i=1}^{\infty} \frac{i}{(i-1)!} = \sum_{i=0}^{\infty} \frac{i+1}{i!} = 2 \sum_{i=0}^{\infty} \frac{1}{i!}.$$

Thus,

$$\lim_{n \rightarrow \infty} \sum_{i=0}^n i(i-2) \binom{n}{k} \left(\frac{1}{n}\right)^i \left(\frac{n-1}{n}\right)^{n-i} = 0.$$

Growth Models

- Many graphs in real world start as small graphs and grow over time.
- There are many ways to attach edges.
- One is to select two vertices uniformly from the set of existing vertices.
- Another is to select two vertices with probability proportional to their degree, which is called **preferential attachment**.
- A variant of preferential attachment is to add a new vertex at each unit of time and add an edge with probability δ , where one end of edge is the new vertex and the other end is a vertex selected according to its degree.
- The graph generated by the last method is a tree with a power law degree distribution.

Growth model without preferential attachment

- We first consider a growth model for a random graph without preferential attachment.
- At each unit time, a new vertex is added and two randomly chosen vertices are linked with probability δ . We allow multiple edges between vertices and the result is **multi-graph**.
- Let $d_k(t)$ be the expectation of the number of vertices of degree k at time t . Then, it satisfies

$$d_0(t+1) = d_0(t) + 1 - 2\delta \frac{d_0(t)}{t}.$$

Here, $\frac{d_0(t)}{t}$ is the expected number of degree 0 vertices that is selected.

Growth model without preferential attachment

- Similarly, we have the following inductive formula for $d_k(t)$:

$$d_k(t+1) = d_k(t) + 2\delta \frac{d_{k-1}(t)}{t} - 2\delta \frac{d_k(t)}{t}.$$

- Suppose there exists a solution of the form $d_k(t) = p_k t$ (asymptotically). Then, we have

$$(t+1)p_0 = p_0 t + 1 - 2\delta p_0, \quad \text{i.e., } p_0 = \frac{1}{1+2\delta},$$

and

$$(t+1)p_k = p_k t + 2\delta p_{k-1} - 2\delta p_k, \quad \text{i.e.,}$$

$$p_k = \frac{2\delta}{1+2\delta} p_{k-1} = \frac{1}{1+2\delta} \left(\frac{2\delta}{1+2\delta} \right)^k.$$

- This implies that the asymptotic degree distribution would be a power law distribution.

The generating function for component size

- Let $n_k(t)$ be the expected number of components of size k at time t .
- As before, we have

$$n_1(t+1) = n_1(t) + 1 - 2\delta \frac{n_1(t)}{t}.$$

- For $k > 1$, based on the same idea as before, we have

$$n_k(t+1) = n_k(t) + \delta \sum_{j=1}^{k-1} \frac{jn_j(t)}{t} \frac{(k-j)n_{k-j}(t)}{t} - 2\delta \frac{kn_k(t)}{t}.$$

Here, $\frac{jn_j(t)}{t}$ is the probability of choosing a vertex in the component of size j .

- We now consider the solution of the form $n_k(t) = a_k t$. After substituting this representation into inductive formula, we have

$$a_1 = \frac{1}{1+2\delta}, \quad a_k = \frac{\delta}{1+2k\delta} \sum_{j=1}^{k-1} j(k-j)a_j a_{k-j}.$$

Generating function

- Let $g(x)$ be the generating function for the component size, where the coefficient of x^k is the probability that a chosen vertex is in a component of size k

$$g(x) = \sum_{k=1}^{\infty} ka_k x^k.$$

- Note that $g(1) = \sum_{k=1}^{\infty} ka_k$ is the probability that a randomly chosen vertex is in a finite sized component.
- It can be shown that g satisfies the following differential equation:

$$g = -2\delta xg' + 2\delta xgg' + x, \quad g(0) = 0.$$

- Heuristically, when $\delta = 0$, there will be no edge and $g(1)$ should be 1. Moreover, as δ increases, $g(1)$, which is the probability that a vertex is in a finite sized component will decrease and in fact, it can be shown that $g(1) < 1$ when $\delta = 1$.

Phase transition for nonfinite component

- Thus, there is a critical value for δ_{crit} such that for $\delta > \delta_{\text{crit}}$, $g(1) < 1$, which implies that the probability of vertex be in a infinite size component is positive.

- Note that

$$g'(x) = \frac{1}{2\delta} \frac{1 - \frac{g(x)}{x}}{1 - g(x)},$$

and whenever, $\delta < \delta_{\text{crit}}$, $g(1) = 1$ and thus,

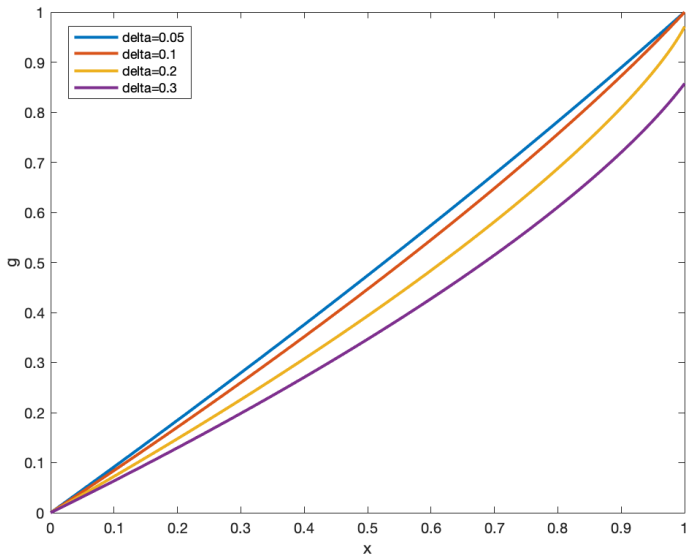
$$\lim_{x \rightarrow 1} g'(x) = \lim_{x \rightarrow 1} \frac{1}{2\delta} \frac{\frac{xg'(x) - g(x)}{x^2}}{g'(x)} \bigg|_{x=1}.$$

- This implies

$$(g'(1))^2 = \frac{1}{2\delta} (g'(1) - 1), \quad g'(1) = \frac{1 - \sqrt{1 - 8\delta}}{4\delta} \text{ (valid one).}$$

- Thus, $\delta = \frac{1}{8}$ is a critical value and this is true from the numerical validation.

The graphs for $g(x)$



Growth model with preferential attachment

- Now, let us consider a growth model with preferential attachment. A vertex is added to the graph for each unit time. The edge is attached to the new vertex and a vertex randomly selected with a probability proportional to its degree.
- Let $d_i(t)$ be the expected degree of the i -th vertex at time t . Then, since the expected number of total edge is δt and the total sum of degree is twice of edge number, the expected value of total degree is $2\delta t$.
- Thus, the probability of i -th vertex being chosen is $\frac{d_i(t)}{2\delta t}$, which implies the following differential equation for d_i :

$$d_i' = \delta \frac{d_i}{2\delta t} = \frac{d_i}{2t} \implies d_i(t) = a\sqrt{t}.$$

- However, at time $t = i$, $d_i(t) = \delta$ (since it should be connected to some vertex with probability δ). Thus, $d_i(t) = \delta\sqrt{\frac{t}{i}}$.

Degree distribution for preferential attachment

- From the formula in the previous slide, we have

$$d_i(t) < d \Leftrightarrow \delta \sqrt{\frac{t}{i}} < d \Leftrightarrow i > \frac{\delta^2}{d^2} t.$$

- Thus, $P(\text{degree} < d) = 1 - \frac{\delta^2}{d^2}$ which implies that the probability distribution $p(d)$ of degree is

$$p(d) = \left(1 - \frac{\delta^2}{d^2}\right)' = \frac{2\delta^2}{d^3},$$

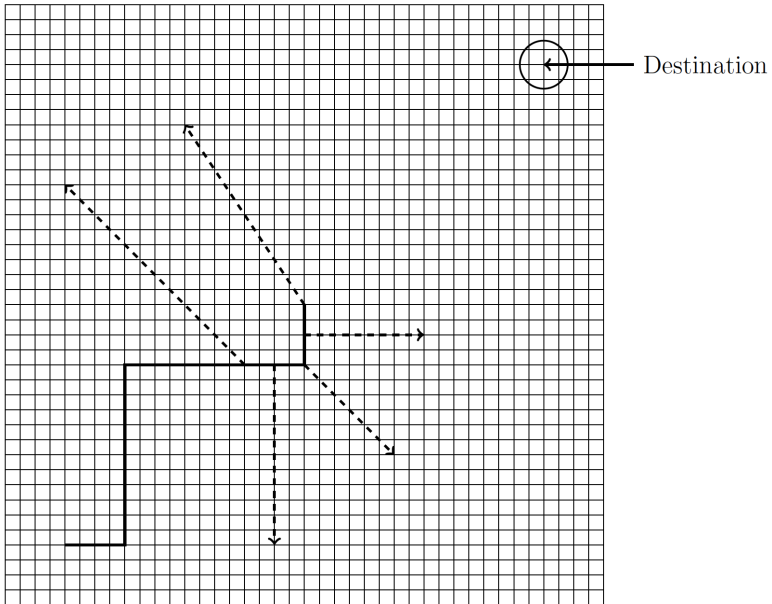
which is power law distribution.

Small World Graphs

Small world graph

- The phrase “Six degrees of separation” implies that any two individuals are connected within six steps.
- This is due to the existence of long-distance edge. In many situations, phenomena are modeled by graphs with local and long-distance edges.
- Consider a 2-dimensional $n \times n$ grid where each vertex is connected to its four adjacent vertices, together with additional long distance edges. Suppose the probability of existence of long-distance edge between vertices u and v is proportional to $(d(u, v))^{-r}$.
- When $r = 0$, the end of the long-distance edge at u is uniformly distributed, and as r increases, the length of long-distance edge decreases. For $r = \infty$, there is no long-distance edge.
- We consider the problem of finding shortest path between two vertices.

Small world graph



- Depending on the parameter r , we have different phenomena.
- When $r < 2$, the end points are too uniformly distributed. Thus, it is unlikely to encounter a long-distance edge whose end point is close to the distance.
- When $r = 2$, there are short paths and there exists an algorithm to find it.
- When $r > 2$, there is no “local algorithm” to find a short path, and indeed, there is no short path at all.
- For given u , let $c_r(u) = \sum_v d^{-r}(u, v)$ be the normalizing constant.

Small world graph

- We estimate $c_r(u)$ according to r .
- For $r > 2$, since there is at least $k < \frac{n}{2}$ vertices with distance k from u , we estimate c_r as

$$c_r(u) = \sum_v d^{-r}(u, v) \geq \sum_{k=1}^{n/2} (k) k^{-r} = \sum_{k=1}^{n/2} k^{1-r} \geq 1.$$

- For $r = 2$, since there is at most $4k$ vertices with distance k from u ,

$$c_r(u) = \sum_v d^{-2}(u, v) \leq \sum_{k=1}^{2n} 4k k^{-2} \leq 4 \sum_{k=1}^{2n} \frac{1}{k} \approx C \log n.$$

- For $r < 2$, similar to $r > 2$ case,

$$c_r(u) = \sum_v d^{-r}(u, v) \geq \sum_{k=1}^{n/2} k^{1-r} \geq \sum_{k=n/4}^{n/2} k^{1-r} \geq C n^{2-r}.$$

No short paths exist for the $r > 2$ case

- We will show that for at least half of the pairs of vertices, there is no short path (polylog steps) between them.
- We first show that there is no edges of length greater than $n^{\frac{r+2}{2r}}$.
- The probability of a particular edge of length greater than $n^{\frac{r+2}{2r}}$ is bounded above by $cn^{-\frac{r+2}{2}}$. Thus, the expected number of edges of length at least $n^{\frac{r+2}{2r}}$ is at most $cn^2 n^{-\frac{r+2}{2}} = cn^{\frac{2-r}{2}} \rightarrow 0$ as $n \rightarrow \infty$.
- Note that for at least half of the pairs of vertices, the distance between them is greater than $\frac{n}{4}$. Since there is no edge longer than $n^{\frac{r+2}{2r}}$, there should be at least $\frac{n}{4} / n^{\frac{r+2}{2r}} = \frac{1}{4} n^{\frac{r-2}{2r}}$ edges between them, which is not polylog.

An algorithm for $r = 2$

- The local algorithm for the case $r = 2$ is, of course, choosing the edge that ends closest to the destination. This algorithm provide the expected length $O((\log n)^3)$. Suppose that the distance between u and v is k . Then, we will show that after $O((\log n)^2)$ steps, we will arrive to a point whose distance to v is at most $k/2$.
- By the triangle inequality, the length between u and any point at distance at most $k/2$ from v is less than $\frac{3k}{2}$. Thus, the probability of existing path from u to those point is at least $Ck^2 \frac{k^{-2}}{\log n} = \frac{C}{\log n}$. Thus, for $O((\log n)^2)$ steps the probability of not existing such a path can be bounded as

$$\left(1 - \frac{C}{\log n}\right)^{c \log n^2} \approx \frac{c}{n} \rightarrow 0.$$

- Thus, after $O(\log n^2)$ steps, the distance from v will be halved and thus, after $O(\log n^3)$ steps, we will arrive to v .

A local algorithm cannot find short paths for $r < 2$

- For simplicity, we only consider the case when $r = 0$. In this case, every end points are equally likely to be connected. Thus, the probability of a long-distance edge hitting one of the n vertices that are within distance $\sqrt{n}$ of the destination is $\frac{O(n)}{n^2} = O\left(\frac{1}{n}\right)$.
- Thus, the probability that the path does not encounter such edge within $\sqrt{n}$ trials is

$$\left(1 - \frac{1}{n}\right)^{\sqrt{n}} \rightarrow 1, \quad \text{as } n \rightarrow \infty.$$

- Since with probability $\frac{1}{2}$, the starting point is at distance at least $\frac{n}{4}$ from destination, and the path will not encounter a long-distance edge ending within distance $\sqrt{n}$ of the destination for $\sqrt{n}$ steps, the expected time is at least $\frac{1}{2}\sqrt{n}$, which is not polylog time.

**THANK YOU VERY MUCH FOR YOUR
ATTENTION.**